LESSON 3.6

Ratio Reasoning: Convert Measurements within the Same System

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratio reasoning to convert between units within the same measurement system.

WHY THIS MATTERS Chef Reyes writes every recipe in the units the supplier uses, but the kitchen's measuring tools are marked in different units of the same system. A stock order lists quarts; the pot is marked in cups. A height rule at the county fair ride is posted in inches; the sign-up sheet lists feet. Your job is to convert between units without changing the amount. The stock order lists 3 quarts, but the pot is marked in cups. The height rule at the ride says 48 inches, but the sign-up sheet lists feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Conversion factor *Spanish: Factor de conversión*	The ratio that links two units in the same measurement system, such as 12 inches to 1 foot. Within one system it is exact.	1 ft = 12 in
Unit ratio *Spanish: Razón unitaria*	A ratio that compares an amount to exactly 1 unit of another quantity, like 12 inches to 1 foot.	12 : 1 — twelve inches for every one foot
Convert *Spanish: Convertir*	To change a measurement from one unit to another without changing how much it is.	6 ft → 72 in
Measurement system *Spanish: Sistema de medidas*	A family of units that go together — customary units like inches and feet, or metric units like milliliters and liters.	Customary: 12 in = 1 ft · Metric: 1,000 mL = 1 L
Double number line *Spanish: Recta numérica doble*	Two number lines lined up so matching tick marks show two units for the same amount.	Liters 0, 1, 2, 3, 4 above milliliters 0, 1,000, 2,000, 3,000, 4,000
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that lets you scale a conversion up or down one row at a time.	1 ft : 12 in, 2 ft : 24 in, 4 ft : 48 in

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A pitcher holds 2 quarts. How many fluid ounces? (1 qt = 4 cups, 1 cup = 8 fl oz)

1. Quarts to cups: 2 × 4 = 8 cups.
2. Cups to fl oz: 8 × 8 = 64 fl oz.
3. Check: 64 ÷ 8 ÷ 4 = 2 quarts.

Answer: 2 quarts = 64 fluid ounces

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Diamond wants to drink 4 liters of water. Her bottle is marked in milliliters.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I know 1 _____ = _____ _____ .
- I _____ because I am going from a _____ unit to a _____ unit.
- My answer is _____ _____ .
- This is reasonable because _____ .

Word bank convert | conversion factor | customary | multiply | divide | larger | smaller | inch | foot | yard | pound | gallon

WATCH OUT!

A common mistake

A common mistake when converting within a measurement system is choosing the operation by habit instead of by unit size

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

There are 12 inches in 1 foot. How many inches are in 5 feet?

(A) 7 inches
(B) 17 inches
(C) 50 inches
(D) 60 inches

STEPS TO TRY
1. Write the unit ratio first: 12 inches : 1 foot.
2. Inches are smaller, so the number gets bigger — multiply.
3. 5 × 12 = ?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

There are 1,000 milliliters in 1 liter. How many milliliters are in 3 liters?

(A) 30 mL
(B) 300 mL
(C) 1,003 mL
(D) 3,000 mL

STEPS TO TRY

1. The unit ratio is 1,000 mL : 1 L.
2. Milliliters are smaller, so multiply.
3. 3 × 1,000 = ?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

There are 16 ounces in 1 pound. How many pounds is 48 ounces?

(A) 3 pounds
(B) 32 pounds
(C) 64 pounds
(D) 768 pounds

STEPS TO TRY

1. Which unit is larger, an ounce or a pound?
2. Converting to the larger unit takes fewer of them, so divide.
3. 48 ÷ 16 = ?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A child is 4 feet tall. A ride requires 44 inches. Can the child ride?

(A) No, because 4 is less than 44

(B) Yes, because 4 feet = 44 inches exactly

(C) Yes, because 4 feet = 48 inches, which is more than 44

(D) You cannot compare feet and inches

STEPS TO TRY

1. You cannot compare two different units directly.
2. Convert the feet to inches with the 12 : 1 unit ratio.
3. Then compare 48 with 44.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does converting to a smaller unit always make the NUMBER bigger, even though the amount stays exactly the same?

SENTENCE STARTER When the unit gets ___, it takes ___ of them to make the same amount, so I ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help